

Release Note

SDK Version	Release Date
v3.12.0.3	2022/10/26
v3.13.0.9-Beta	2023/05/05
v3.13.0.16	2023/06/09
v3.13.0.26	2023/09/08
v3.13.0.40	2023/11/10
v3.13.0.41	2023/12/27
v3.13.0.60	2024/02/02
v3.13.0.79	2024/05/29
v3.13.0.80	2024/06/27
v3.13.0.86	2024/10/22
v3.13.0.90	2025/02/13
v3.13.0.91	2025/09/29
v3.13.0.95	2026/05/28

Version: v3.13.0.95

Changes

- Optimized SDK sample code and documentation for better usability and integration experience.
 - Updated copyright and license information.
 - Upgraded mbedtls to version 3.6.4 with refreshed security configurations.
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New Features

- **APP-bt_audio_trx:**
 - Enhanced Find My integration, including GFPS TX power configuration and modularized Find My / GFPS code.
 - Added LE Audio voice and recording EQ features, and optimized local playback and audio stream control.
 - Improved COD configuration and refined several app event names for better readability and maintainability.
 - Added external PA support and mSBC/CVSD codec conversion capability.
-

Fixed Issues

- Fix most of issues reported in previous SDK
-

Known Issues

1. `bt_audio_trx` transmitter

1.1 Pop noise at the headphone side during Line-in + HFP use. When using `bt_audio_transmitter` with Line-in and HFP concurrently, a short "pop" noise may be heard at the headphone side during call.

2. `bt_audio_trx` receiver

2.1 No audio output after long-duration MCP playback. During long-duration MCP playback using `bt_audio_receiver`, audio output may stop even though the media session remains active.

SDK API Change List

- See SDK documentation for detailed API changes
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Notes

1. Important Notes

- GATT over BR/EDR is now disabled by default. To enable it, set `F_APP_GATT_OVER_BREDR_SUPPORT` in `app_flags.h`.
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Version: v3.13.0.91

Changes

- Optimize documents by correcting some formatting and spelling errors in on-line documents;
- Modify the naming of some documents and tools according to the new naming rules.

New Features

1. Add `AI Summary` and `AI Ask` functions to on-line documents;

Fixed Issues

1. None

Known Issues

1. None

SDK API Change List

1. None

Notes

1. Important attentions of this version

- None
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Version: v3.13.0.90

Changes

- Optimize SDK sample code
- Optimize SDK documents, and the format of documents are changed to `RST (reStructuredText)`

New Features

1. Add new part number `RTL8763ESE-VI` and `RTL8763EAV-VI`
2. More features are added to `bt_audio_trx`
 - Optimize some code flow
 - Add `F_APP_SD_CARD_SUPPORT` macro to separate sd card related code
 - Add attach local play for srcplay module

Fixed Issues

1. Fix most of issues reported in previous SDK
2. Fix some GATT over BR/EDR IOP issues with Phones

Known Issues

1. `bt_audio_trx`
 - 1.1 After DUT reset, find my accessory fails to reconnect successfully.
 - 1.2 When adjusting the A2DP EQ with only the LE link present, a hard fault occurs.

SDK API Change List

see in SDK doc

Notes

1. Important attentions of this version

- The default bin of the `bt_audio_receiver` GCC version will hard fault due to insufficient RAM upon startup. If customers need to verify the GCC version of `bt_audio_receiver`, they need to recompile the app themselves. We have addressed this issue by disabling some modules in the source code `app_flags.h`.
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Version: v3.13.0.86

Changes

- Optimize SDK sample code
- Move file `\inc\rt187x3e\hal\rt1876x_captouch.h` to `\inc\rt187x3e\io\rt1876x_captouch.h`
- Move file `\inc\rt187x3e\hal\rt1876x_sleep_led.h` to `\inc\rt187x3e\io\rt1876x_sleep_led.h`
- Move file `\inc\app\app_timer.h` to `\inc\hal\platform\app_timer.h`
- Remove unused files `app_types.h`, `engage.h`, `legacy_gaming.h` in `\inc\app\`

New Features

1. Add GATT over BR/EDR support
2. Add new part number `RTL8763EWS`
3. Add `\src\ble\bt_gatt_client\ancs_gatt_client.c/h`
4. Add `\src\ble\bt_gatt_client\gaps_gatt_client.c/h`
5. Add touch `chsc6417` and lcd `sh8601z_454_spi` , `st7801_368_448_qspi` driver
6. Add BT PAN demo support
7. Add integration with lwIP support
8. Add integration with mbedTLS support
9. More features are added to `bt_audio_trx`
 - GATT over BR/EDR,
 - HOGP Host, the code for HOGP is optimized into two parts: HOGP Host and HOGP Device
 - Add an application scenario that can support HFP AG/HF or A2DP Source/Sink on a single chip, please refer to the code included in the macro definition `F_APP_INTEGRATED_TRANSCEIVER`
 - Rename `F_APP_BT_AUDIO_USB_DEMO_SUPPORT` to `F_APP_USB_AUDIO_SUPPORT`
 - Add BR/EDR Secure Connection and CTKD

Fixed Issues

1. Fix most of issues reported in previous SDK

Known Issues

1. `bt_audio_trx`
 - 1.1 After DUT reset, find my accessory fails to reconnect successfully.
 - 1.2 When the 2 one-wire UART function is enabled, there is no response from the earphone end after the DUT is powered on.
 - 1.3 In this application scenario that can support HFP AG/HF or A2DP Source/Sink on a single chip, during a call, if the distance is increased, the Bluetooth link will disconnect. When the distance is reduced, both ends of the call will be silent.

SDK API Change List

see in SDK doc

Notes

1. Important attentions of this version

- GATT over BR/EDR functionality is enabled by default through the macro definition `F_APP_GATT_OVER_BREDR_SUPPORT`

Version: v3.13.0.80

Changes

- Optimize SDK sample code
- Optimize i2s code

New Features

Fixed Issues

1. Fix most of issues reported in previous SDK

Known Issues

1. `bt_audio_trx`
 - 1.1 Occasional LE link disconnection during stress test.

SDK API Change List

see in SDK doc

Notes

1. Important attentions of this version

Version: v3.13.0.79

Changes

- Optimize SDK sample code
- Some default features in "`bt_audio_trx`" have been subtly adjusted. Some features that were enabled in the previous version of the SDK might be disabled in the current version, while some features that were disabled in the previous version might be enabled in the current version. Please compare the differences in "`app_flags.h`" to see these changes.

- The bin of OTA Header is moved to directory `BANK0\OTAHeader\gcc` or `mdk`
- Remove `APP_` prefix for `APP_EVENT_XXX`, and the file `app_report_common_sdk.h` is removed.
- Rename `F_APP_SPP_CAPTURE_DSP_DATA_2` to `F_APP_DATA_CAPTURE_SUPPORT`

New Features

1. Add 1M flash map for bt hid demo
2. Add document `PTS Test User Guide`
3. Add new part number `RTL8773ESL`
4. Fix UI stuck and add TE support for GUI
5. Add GATT client database hash support
6. Add touch `cst816d` and LCD `st77916` driver support
7. More features are added to `bt_audio_trx`
 - Disable notification support for transceiver demo(`F_APP_BT_AUDIO_TRANSCEIVER_DEMO_SUPPORT`)
 - Add VCP/VCS support

Fixed Issues

1. Fix most of issues reported in previous SDK

Known Issues

1. `bt_audio_trx`
 - 1.1 GFPS finder can't work
 - 1.2 When connected to a PC, test with the `bt_audio_receiver`, there is no sound when receiving incoming calls via Teams.

SDK API Change List

see in SDK doc

Notes

1. Important attentions of this version

Version: v3.13.0.60

Changes

- Optimize SDK sample code
- Refactor usb code
- Add bt/ble prefix for some bt/ble demo project, which is listed in table below:

Old name	New name
iap_demo	bt_iap_demo
pbap_demo	bt_pbap_demo
spp_demo	bt_spp_demo
dtm	ble_dtm

- Move `rtl87x3e_bt_audio_trx.uvprojx` from `sdk\board\evb\bt_audio_trx` to `sdk\board\evb\bt_audio_trx\mdk`
- Update VP resource, add more Chinese VP resource
- Add `CONFIG_REALTEK_` prefix for gcc support
- Remove some used code, such as OTA tooling, gcsc and gcscs

New Features

1. Add gui demo and lcdc demo project
2. Add new part number `RTL8763EWE` and `RTL8763EWM`
3. More features are added to `bt_audio_trx`
 - Add uart dfu support
 - Add acoustic mp support
 - Add GPIO/MFB key support
 - Add 16M target for each demo project
 - Add usb mass storage support
 - Add MP3 decode support for ULL dsp bin
 - Add support bis and cis simultaneously in `bt_audio_receiver` demo bin
 - Add BR/EDR secure connection + CTKD support
 - Add UAC + SCO/CIS bi-direction audio support with ULL DSP bin
 - Add USB CFU support
 - Add `F_APP_CHARGE_CASE_DEMO_SUPPORT` in `app_flags.h` to support charging case demo
 - Add `F_APP_DASHBOARD_DEMO_SUPPORT` in `app_flags.h` to support dashboard demo
4. Add dis client support

Fixed Issues

1. Fix most of issues reported in previous SDK

Known Issues

1. bt_audio_trx
 - 1.1 Occasional LE link disconnection during stress test.
 - 1.2 System can't enter power down mode if GUI is enabled.
2. bt_audio_receiver
 - 2.1 CIS: Occasional audio stuck when receive audio stream.
 - 2.2 BIS: Can't work with Auracast assistant.

SDK API Change List

see in SDK doc

Notes

1. Important attentions of this version

Version: v3.13.0.41

Changes

1. Remove the beta version of the GUI design tool `RVisualDesigner-v1.0.4.3.zip` and the corresponding documentation `RTKIoT Visual Designer User Guide EN.pdf` from SDK package; it is still on development, SDK will support this feature in later release.

New Features

None

Fixed Issues

None

Known Issues

None

SDK API Change List

None

Notes

1. Important attentions of this version

None

Version: v3.13.0.40

Changes

1. Optimize SDK sample code
2. Remove prefixes `APP_` from macro definition names `APP_CMD_XXX`
3. File directory `target`, which includes four files are removed in some sample projects, such as `board\evb\ble_bt5_central\target`
4. Remove default `*.rcfg` for external DSP
(`RTL8763xxx_CommonSDK_external_dsp_fixed_48K_master_sample.rcfg`), you can configure Audio Route to use external DSP via MCUConfig tool

New Features

1. Add new part number `RTL8763EW-VC` and `RTL8763EWE-VP`
2. Add 16M flash map for `RTL8763EW-VC` and `RTL8763EWE-VP`
3. Add GUI(beta version) support, a new directory `src\sample\gui` was added
4. Add BT MAP demo project
5. Add BT PBAP demo project
6. Add core dump parser, please refer to the document `Debug Analyzer User Guide` for details
7. Add iOS and Android audio connect SDK
8. Add ams, ancs, and ble hid client for new gatt interface
9. More features are added to `bt_audio_trx`
 - Add new target `bt_audio_trx_16M_1ea_dual_bank0/1` for new part number `RTL8763EW-VC` and `RTL8763EWE-VP`
 - Add GUI support (default capacitive touch: `CST816T`, Display: `SH8601Z`)
 - Add OTA support for USER DATA
 - Add USB audio source in support
 - Add local sd card mp3 source playback
 - Add one wire UART
 - App GFPS finder
 - Add CIS and CIS ULL support
 - Add Auracast Assistant support
 - Add DLPS support and DLPS is enabled by default
 - Optimize app cfg module, separate the app cfg into sub-modules
 - Move the pinmux definition to the `board.h`
10. Add more documents in `doc/EN`
 - Android Audio Connect App User Guide EN.pdf
 - iOS OTA APP User Manual EN.pdf

- Image Convert Tool User Guide EN.pdf
- RTKIOT Visual Designer User Guide EN.pdf
- RTL87x3 Current Measure Guide EN.pdf
- RTL87x3E Audio Plugin User Manual EN.pdf
- RTL87x3E Audio Route User Manual EN.pdf
- RTL87x3E Audio Subsystem Introduction EN.pdf
- RTL87x3E BREDR Stack User Manual EN.pdf
- RTL87x3E BT Audio Transceiver Design Spec EN.pdf
- RTL87x3E Platform Power Test User Manual EN.pdf
- RTL87x3E BT MAP Demo Application Note EN.pdf
- RTL87x3E PBAP Demo Application Note EN.pdf
- RTL87x3E A2DP Application Note EN.pdf
- RTL87x3E AVRCP Application Note EN.pdf
- RTL87x3E HFP Application Note EN.pdf
- RTL87x3E PBAP Application Note EN.pdf
- RTL87x3E SPP Application Note EN.pdf
- RTL87x3E iAP Application Note EN.pdf
- RTL87x3E OPP Application Note EN.pdf

11. Add sidebar for each Application Notes and User manual in `BT-AUDIO-SDK.chm`

Fixed Issues

1. Fix most of issues reported in last SDK v3.13.0.26

Known Issues

1. `bt_audio_trx`
 - 1.1 Occasional LE link disconnection during stress test.
 - 1.2 System can't enter power down mode if GUI is enabled.

SDK API Change List

see in SDK doc

Notes

1. Important attentions of this version

- To support the 16M flash map, some adjustments have been made to the file directory `bin\rtl87x3e\app_demo_bin\bt_audio_trx`, and two folders `4M` and `16M` were added. For `RTL8763ESE/EFL/EAU`, it should be `4M`, for `RTL8763EW-VC/EWE-VP`, it should be `16M`
- `default_rsa_key` shall be burned to chip `RTL8763EW-VC/EWE-VP`, otherwise the chip will fail to boot.
- The default configuration of the LE link has been modified, only 1 slave link is configured. The `RTK OTA` advertisement will start automatically when the BLE link is connected. if you want to enable more BLE

advertisements, please disable the `RTK_OTA` advertisement by yourself by modifying the code or changing the `LE_slave_link` in the MCUConfig tool to 2 or more.

- More and more features are supported in `bt_audio_trx`, but memory space is limited. Enabling more functions may result in insufficient memory, please make sure that there is enough memory when opening more features in `app_flags.h`, otherwise it may cause the system to fail to boot or hard fault.
 - Some features are not enabled by default in `app_flags.h`, such as SD card playback and SD card + BIS/CIS transmitter in the `bt_audio_trx` demo.
-

Version: v3.13.0.26

Changes

1. Optimize SDK sample code
2. Remove prefixes `APP_` from macro definition names `APP_CMD_XXX`

New Features

1. Add BIS ULL(Ultra low latency) support
2. Add Apple FindMy support
3. Add DSP spp data capture support
4. Add new document `RTL87x3E Qualification User Manual EN` and `RTL87x3E SoC Platform Tutorial EN`

Fixed Issues

1. Fix most of issues reported in last SDK v3.13.0.16

Known Issues

1. `bt_audio_trx`
 - 1.1 A2DP Transparent Transmission
 - Audio Stuttering occasionally when play music;
 - 1.2 line in audio from external dsp can't work with dsp image `cs_pipe_stereo`, but it works well with dsp image `cs_1ea_pipe_stereo`;
2. `audio_demo`
 - 2.1 linein audio can't work with dsp image `cs_pipe_stereo`, but it works well with dsp image `cs_1ea_pipe_stereo`;

SDK API Change List

see in SDK doc

Notes

1. Important attentions of this version

- Some LE Audio features have been added to `bt_audio_trx`, but the functionality is not fully verified yet.
 - Due to lack of ram, voice speaker algorithm are removed(modules included in the red box below) in `cs_pipe_stereo` DSP image.
-

Version: v3.13.0.16

Changes

1. Optimize SDK sample code

New Features

1. Add OTA(over SPP and GATT) support in bt_audio_trx
2. Add BT HCI snoop log capture support in DebugAnalyzer tool
3. Add device name configuration in MCU Config Tool
4. Rename hid_demo to bt_hid_demo
5. Add SPI external flash demo support in io_demo project
6. Add power test bin

Fixed Issues

1. Fix most of issues reported in last SDK v3.13.0.9-Beta
2. Fix Factory reset and Device name setting not work issue in previous SDK

Known Issues

1. bt_audio_trx
 - 1.1 A2DP Transparent Transmission
 - Audio Stuttering occasionally when play music;

SDK API Change List

see in SDK doc

Notes

1. Important attentions of this version

Version: v3.13.0.9-Beta

Changes

New Features

1. Add line-in support in audio demo
2. Add BT hid device demo
3. Add iap demo
4. Add BLE audio samples demo
5. Add BT audio transceiver (bt_audio_trx) demo
6. Add BIS support in bt_audio_trx;
7. Add I2S interface support in SDK;

Fixed Issues

1. Fix all issues reported in last SDK v3.12.0.3.

Known Issues

1. usb_audio demo
 - 1.1 usb audio can't work.
2. bt_audio_trx
 - 2.1 HFP Transparent Transmission
 1. During hfp transparent transmission, take headset out of ear, noise heard in phone side, it can't recover when put headset in ear, however it can recover if you input cmd "hfpag conn" to Device 2;
 - 2.2 A2DP Transparent Transmission
 1. Audio Stuttering occasionally when play music;
 - 2.3 Source Play function
 1. i2s and mic source play function can't work if sample rate of audio track is 48K.
 - 2.4 Record function
 1. There is a memory leak in the PCM record function, which will cause hard fault.
 - 2.5 Occasional VP does not work properly.
3. ota_demo
 1. Tested with some phones and found that OTA upgrade is too slow.

SDK API Change List

see in SDK doc

Notes

1. Important attentions of this version

1. ANC and APT are not supported in Common SDK (audio_demo doesn't support ANC and APT either).
2. Operating environment of app demo bins

	RTL8763ESE	RTL8763EAU	RTL8763EFL
audio_demo	Y	Y	Y
ble_broadcaster	Y	Y	Y
ble_observer	Y	Y	Y
ble_peripheral	Y	Y	Y
ble_central	Y	Y	Y
ble_scatternet	Y	Y	Y
ble_bt5_peripheral	Y	Y	Y
ble_bt5_central	Y	Y	Y
bt_audio	Y	Y	Y
bt_audio_trx	Y	Y	Y
bt_hid	Y	Y	Y
dtm	Y	Y	Y
iap_demo	Y	Y	Y
lea_cap_acceptor	Unsupported	Y	Y
lea_cap_commander	Unsupported	Y	Y
lea_cap_initiator	Unsupported	Y	Y
ota_demo	Y	Y	Y
spp_demo	Y	Y	Y
usb_audio	Y	Y	Unsupported
usb_hid	Y	Y	Unsupported

3. iap_demo

1. The test of this demo needs user to install iOS audio connect tool, which is not formally released yet. Customer can write its own iOS app which supports iAP to test with iap_demo.

4. bt_adudio_trx

1. Three documents are provided for bt_adudio_trx
 - a) RTL87X3E BT Audio Transceiver Application Note EN.pdf

b) RTL87X3E BT Audio Transceiver ACIHostCLI User Manual EN.pdf

c) RTL87X3E BT Audio Transceiver ACI User Manual EN.pdf

You can also refer these documents from BT-AUDIO-SDK.chm

2. Audio control such as play, pause, etc. are not supported in a2dp transparent transmission yet

3. Voice control such as answer, reject, mute, unmute, etc. are not supported in HFP Transparent Transmission

4. All supported functions are listed in doc RTL87X3E BT Audio Transceiver Application Note EN.pdf

5. DSP images

There are 2 types of images for RTL8763ESE, RTL8763EAU and RTL8763EFL. We just take RTL8763ESE as an example;

- o cs: common sdk
- o pipe: audio pipe
- o lea: le audio

DSP Image Type	Directory	Function List
cs_pipe_stereo	bin\rtl87x3e\8763ESE\BANK0	Audio(A2DP) Voice(HFP, 1 mic) Linein Audio Pipe MP3 decoder DSP share 80KB SRAM to MCU
cs_lea_pipe_stereo	bin\rtl87x3e\8763ESE\DSP_Image\cs_leaaudio_pipe_stereo	Audio(A2DP) Voice(HFP, 1 mic) Linein Audio Pipe MP3 decoder LE Audio

Version: v3.12.0.3

Changes

New Features

1. First version of Common SDK

2. Support rich demo apps

audio_demo

bt_audio

io_demo

ota_demo

spp_demo

ble samples

ble_broadcaster

ble_bt5_central,

ble_bt5_peripheral

ble_central

ble_observer

ble_peripheral

ble_scatternet

usb_audio

usb_hid

dtm

Fixed Issues

Known Issues

1. ota_demo

1. Reset DUT, connects BT/BLE from the setting UI(Android or iOS), can't establish connection;

Note: If connect to DUT directly via OTA APP UI, connection can establish successfully, and OTA successfully.

SDK API Change List

see in SDK doc

Notes

1. Important attentions of this version